

LAKE GRACE, Australia

WMO: 956370

Lat: 33.1006S

Lon: 118.4647E

Elev: 940

StdP: 14.20

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	35.6	37.8	22.5	17.6	79.5	26.3	21.0	74.6	24.0	58.9	21.2	59.1	1.8	0	0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.2	99.2	65.1	95.2	65.0	91.6	64.0	78.7	80.1	71.0	80.7	67.5	81.3	11.4	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.4	152.6	79.6	68.8	109.6	72.3	63.9	92.1	70.5	43.0	79.9	35.5	81.4	32.5	81.8	98.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.6	19.5	17.5	31.6	106.9	1.8	2.6	30.3	108.8	29.2	110.3	28.2	111.8	26.9	113.7		
(5)			31.1	85.1	1.6	10.4	29.9	92.6	29.0	98.7	28.1	104.6	26.9	112.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.0	73.0	73.0	69.8	64.1	57.4	52.8	50.7	51.9	54.6	60.8	66.6	69.9
	DBStd	9.77	6.16	5.97	6.35	5.84	4.71	3.91	3.71	4.11	5.23	6.02	6.64	6.52
	HDD50	93	0	0	0	0	2	15	37	25	13	1	0	0
	HDD65	2130	6	6	24	84	241	366	442	406	314	161	59	21
	CDD50	4473	714	644	613	424	231	99	60	85	152	335	498	618
	CDD65	1034	255	230	172	58	5	0	0	3	30	107	174	
	CDH74	12336	3035	2523	1763	543	47	1	0	6	82	531	1472	2333
	CDH80	5827	1594	1255	764	155	5	0	0	0	16	188	662	1188
Wind	WSAvg	9.0	10.9	10.5	9.5	7.9	7.2	7.8	7.7	7.8	8.6	9.6	10.6	10.7
Precipitation	PrecAvg	13.6	0.7	0.7	0.9	0.9	1.7	1.9	1.8	1.6	1.2	0.8	0.9	0.5
	PrecMax	21.5	6.6	3.2	6.0	4.2	5.9	4.4	4.2	4.0	2.5	3.0	5.1	2.7
	PrecMin	7.7	0.0	0.0	0.0	0.0	0.1	0.3	0.5	0.2	0.2	0.1	0.0	0.0
	PrecStd	3.1	1.5	0.8	1.3	0.9	1.2	0.9	0.8	0.8	0.6	0.6	0.9	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	104.2	102.2	98.6	90.9	78.8	70.7	67.0	72.9	82.1	90.9	97.7	103.7
		MCWB	66.9	67.4	67.7	62.8	56.9	55.3	54.5	55.9	57.5	59.9	64.3	63.9
	2%	DB	98.7	96.6	93.4	84.3	74.4	66.8	63.7	67.8	75.4	85.7	92.6	97.3
		MCWB	65.0	65.7	66.6	62.6	56.8	54.0	53.4	54.4	56.0	60.9	62.1	63.7
	5%	DB	93.8	92.6	88.4	80.1	70.7	64.0	61.2	64.4	70.6	79.9	87.9	91.8
		MCWB	65.1	66.2	64.2	61.3	56.0	53.3	53.0	53.9	55.7	59.2	62.3	62.6
	10%	DB	89.1	87.8	83.4	76.2	67.4	61.3	59.0	61.5	66.5	75.0	82.6	86.5
		MCWB	64.2	64.5	63.5	60.0	55.6	53.0	52.1	52.9	54.3	57.2	60.2	62.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	84.1	84.0	85.1	78.8	65.9	62.3	60.6	62.6	66.9	75.6	84.9	84.5
		MCDB	85.2	85.0	85.9	79.4	69.5	62.6	61.9	63.5	67.7	76.2	86.3	86.2
	2%	WB	72.1	72.5	70.8	66.0	61.3	58.4	56.7	58.1	59.9	63.1	68.0	68.1
		MCDB	84.6	82.3	75.7	75.7	67.7	62.1	59.9	63.1	69.7	76.3	72.5	84.5
	5%	WB	68.4	68.5	66.9	63.5	59.1	56.2	54.6	55.9	57.3	60.3	63.4	65.0
		MCDB	84.3	83.4	79.4	74.3	65.9	60.6	58.5	61.7	67.3	75.2	80.0	83.8
	10%	WB	65.8	66.3	65.0	61.6	57.2	54.4	52.8	53.9	55.4	58.3	61.3	63.0
		MCDB	84.1	82.2	77.9	71.8	64.0	58.9	57.5	59.5	64.2	72.1	78.2	81.8
Mean Daily Temperature Range	5% DB	MDBR	27.2	25.6	23.1	20.6	19.3	17.4	16.9	18.7	21.9	24.4	26.2	27.3
		MCDBR	35.2	33.2	29.7	26.4	24.3	21.0	19.6	24.1	29.1	32.8	34.7	36.6
		MCWBR	10.6	10.1	10.9	10.8	11.3	11.6	11.9	13.5	13.3	13.1	12.4	10.9
	5% WB	MCDBR	28.7	27.5	23.5	21.4	17.4	16.3	16.4	18.6	23.9	27.3	28.6	30.5
		MCWBR	14.8	14.0	13.3	12.3	10.8	12.2	12.7	13.0	14.2	13.4	15.4	13.3
Clear-Sky Solar Irradiance	taub	0.336	0.330	0.316	0.305	0.291	0.285	0.286	0.290	0.310	0.318	0.322	0.325	
	taud	2.504	2.538	2.562	2.588	2.601	2.598	2.589	2.576	2.493	2.494	2.504	2.512	
	Ebn at Noon	318	313	307	293	282	275	281	294	304	313	320	322	
	Edh at Noon	36	34	31	27	23	22	23	26	32	35	36	36	
All-Sky Solar Radiation	RadAvg	2529	2250	1836	1348	1004	836	887	1145	1544	2003	2351	2589	
	RadStd	150	117	99	85	67	39	49	49	67	113	139	140	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-4.09	-7.24	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page